# Bigger, Longer, Not Cheaper: Price Floors in China's Aggregate Mining Rights Market[*]

Ze Wang · Research proposal · October 25, 2026

## 1. Question

Delegating the allocation of natural resources creates a tension between using local information and protecting public value. Local officials are well placed to read market conditions and design transactions, but their objectives may differ from those of the public owner (Alonso and Matouschek 2008; Duflo et al. 2018). Higher-level governments therefore limit local discretion, and price floors are a common instrument: public assets are often sold behind reserve prices whose minimum is set from above. A floor that blocks undesirable concessions in normal times may block beneficial adjustment when demand falls, yet falling demand may also strengthen local incentives to sell cheaply. Its effect also depends on which decisions remain in local hands: a seller that cannot lower the price may change *what* it sells instead.

This study asks: *when higher-level rules constrain one dimension of local discretion, how do local governments use the discretion that remains to respond to demand shocks, and with what consequences for the allocation of natural resources?* I study China's mining rights for construction aggregates, where provinces set price floors. I plan to compare local sellers facing tighter and looser floors, and larger and smaller declines in demand. The question is whether, when the price is tightly bound, they adjust the scale, duration and timing of the rights they sell instead, and how these adjustments change who obtains the resources and when they are developed. To answer it, I expect to use the staggered revisions of provincial benchmark prices, which tighten or loosen the floor at different times across provinces, together with differences in local exposure to the property-led collapse in construction demand.

## 2. Institutional background

Since the 2017 reform of mineral-rights revenue collection, provinces have set benchmark prices per unit of reserves, meant both to protect the state's ownership interest and to reflect market conditions. These provincial price standards bound how low local governments may price aggregate mining rights, and they applied throughout the 2020–2025 period I study. China's 2023 collection rules kept this system and require benchmarks to be updated, in principle every three years. Provinces revise their schedules on their own timetables, and this staggered timing is the variation my design will use. Within these standards, county governments decide which deposits to offer and when, and propose their scale and duration subject to planning and approval. Over the same period, construction demand collapsed with the property sector: housing starts fell 63% between 2021 and 2024 (NBS 2021, 2024), and national prices of manufactured sand and crushed stone fell by more than a fifth.

## 3. Motivating facts

A cleaned nationwide ledger of 4,742 aggregate mining-right sales in 2020–2025 lets me follow both the prices at which rights were sold and what was sold.[1] Comparing 2024–25 with 2020–21 within prefecture and mineral type reveals a clear tension. While national prices of manufactured sand and crushed stone fell, the size-adjusted announced reserve price per tonne rose by about a third, and the winning price by about a fifth (Figure 1a). More rights sold exactly at their announced starting price: the share rose by 11 percentage points from about half (Figure 1b). At the same time, what was sold changed. Resources per right rose by two-thirds and duration by one and a half years, while the number of sales barely fell (Figure 1c). The price did not follow the market, but the contract did. This points to local adjustment shifting from price toward the scale and duration of the rights on offer.

## 4. Research design

*What adjusts when the price is constrained?* I plan to use a **staggered difference-in-differences design**[2] that compares provinces revising their benchmarks in a given period with provinces that have not yet revised. Treated and comparison units will be drawn from the same mineral categories

[*]This proposal describes a preliminary research idea; all figures are preliminary and subject to revision.

[1]Ccement Hangqingtong announcements reconciled with official result notices (Ministry of Natural Resources and provincial platforms), official values prevailing; product prices are Ccement national indices.

[2]Revision timing may itself respond to local market or fiscal conditions; establishing why and when each revision occurred is the first priority of the next stage.

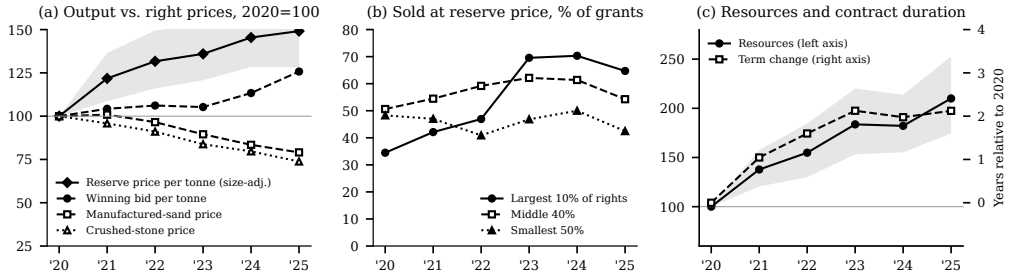



**Figure 1.** Annual estimates with prefecture and mineral fixed effects; prices and resources indexed to 2020=100, duration in years. Reserve prices also control for log resources, duration, log area and a volume-unit indicator. Panel (b) groups rights by pooled reserve-price rank. Shading: prefecture-clustered 95% intervals.

and matched on pre-revision constraints and listing trends. A county facing a tighter floor in a weak market cannot cut the price, but it can adjust scale or duration, potentially improving project viability through economies of scale or greater flexibility in development, or it can hold back offers until conditions improve. I will therefore examine how each revision moves the reserve price together with the size, duration and timing of what counties put on the market, both for individual listings and for the flow of supply in each county. I will track unsuccessful and repeated listings to distinguish changes in offered contracts from changes in which rights successfully sell.

*Why does the price not fall?* Three explanations could produce the same pattern, and each leaves a different footprint. (1) If the floor binds, announced reserve prices should sit at the benchmark-implied floor, move with it when it is revised, and contract terms should change most where the floor bites hardest. (2) If fiscal incentives drive the pattern, counties that depended more on land-sale revenue before the downturn should hold prices above the floor even where it is slack, and enlarge or lengthen rights to sustain receipts. (3) If political accountability drives it, officials should avoid visible price cuts even when the rules allow them, and more so when audits or anti-corruption inspections are active or when the county leaders who run the sales are near promotion. By compiling every provincial benchmark edition and matching each listing to the floor in force when it was offered, I will measure how far announced prices sit above the floor and how this gap varies with fiscal dependence and political exposure.

*How does allocation change?* Whatever the reason, changing what is sold also changes who obtains the resource and when it is used. A larger, longer right may help a sale go through, but it raises the capital needed to bid, which may shift wins toward large or state-owned firms; and it may also delay extraction or leave a site stripped but idle, with environmental damage and little output. I will therefore follow each right from sale to the ground: who wins it, and, from Sentinel-2 imagery, whether its site is worked or left idle.

## 5. Contribution

The main contribution of this project is to show that a higher-level price floor does not stop a public seller from adjusting to a collapse in demand, but redirects the adjustment from price into the design of the asset it sells, and to identify why the price does not fall. Work on rules and discretion studies governments mainly as buyers, regulators or grant-makers, and asks which institutional features make discretion valuable (Bosio et al. 2022; Bandiera et al. 2021; Decarolis et al. 2025; Cingano et al. 2025), building on the theory of optimal delegation (Alonso and Matouschek 2008). This project adds the state of the market: the same floor can be slack in a boom and bind in a bust, so its protective value and its cost to adaptation move with demand. Evidence that officials escape rules through other contract terms comes from public buyers who shift contract values around regulatory thresholds (Coviello et al. 2026; Carril 2026; Szucs 2024). Here a public seller facing a price floor changes the physical scale, duration and timing of what it sells. Work on Chinese local governments as land sellers shows that fiscal motives shape both which land they supply and how they support residential land prices in downturns (He et al. 2023; Chang, Wang and Xiong 2026), but there the government sets the price itself. Because the floor here is set from above and observed, a binding rule, fiscal incentives and political accountability can be separated as reasons why prices do not fall. Finally, because aggregate mining rights are long-lived, the redirected adjustment shapes who obtains natural resources and how they are extracted, linking work on reserve prices and contract duration (Andreyanov and Caoui 2022; MacKay 2022; Noguchi and Otani 2025) to the environmental footprint of extraction.

### a. Background of proposed research plan

How can essential network services such as safe drinking water reach the "last mile" in developing countries? In 2024, 2.1 billion people still lacked safely managed drinking water, with rural areas continuing to lag behind cities (WHO and UNICEF, 2025). The remaining unserved households are disproportionately located in remote, sparsely populated, and low-income communities, where network construction requires high upfront investment despite limited demand and ability to pay. Public budgets and administrative capacity are therefore often insufficient for governments to build and operate every local system directly.

Where public agencies cannot finance and directly operate numerous small, dispersed systems, governments may rely on local private operators to reach underserved communities. Local firms can supplement constrained public finance and managerial capacity and may be better positioned to adapt service delivery to local conditions; World Bank evidence from small and medium-sized towns highlights these limitations of decentralized public provision (Triche, Requena, and Kariuki, 2006). One way to organize such participation is to grant a selected operator exclusive responsibility for a defined service area, while the government regulates tariffs= and coverage. This arrangement can make long-lived infrastructure investment more credible and establish clear responsibility for network expansion. Yet exclusivity also creates an allocation risk: if an operator's financial or technical capacity does not match the spatial structure of its territory, households may remain uncovered while nearby or more capable providers are unable to enter (World Bank, n.d.).

Cambodia exemplifies this broader last-mile challenge: the country relies extensively on private operators to extend piped-water networks into smaller towns and rural communities. Private operators provide over half of the country's piped-water connections, whereas public utilities are concentrated mainly in urban centers (JICA, 2023; World Bank, 2023). As illustrated in Figure 1, the government assigns a geographically defined service area to a selected operator through an area license. The operator finances and builds the water facilities and subsequently operates and maintains the system and collects regulated tariffs, while the government approves supervises the service. Because the license grants exclusive supply rights within the assigned territory, each assignment creates a local monopoly (Royal Government of Cambodia; DFDL, 2023). In practice, service area are designated at the commune and village levels. The spatial allocation of these local monopolies is therefore a central policy choice: if service boundaries shaped partly by administrative geography do not coincide with the lowest-cost configuration of the physical network, they may create hidden costs in last-mile water expansion.

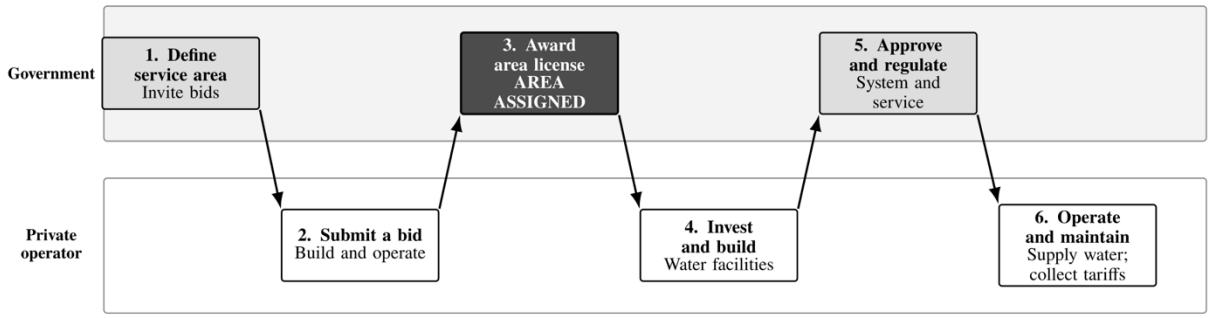



Figure 1Piped-Water Provision in Cambodia.

### b. Purpose of proposed research

This study asks whether Cambodia's assignment of exclusive water-service territories matches villages to the lowest-cost feasible operator, or whether licensing boundaries shaped by administrative geography distort this allocation and raise the cost of piped-water expansion. As illustrated in Figure 2, geographically proximate villages may lie on opposite sides of administrative boundaries that structure licensing assignments, while one village is covered by piped water and the other remains unserved. Although such differences do not by themselves demonstrate inefficiency, they raise the possibility that regulatory service areas do not coincide with the most cost-effective organization of water infrastructure. This concern is also reflected in the 2020 Provincial Investment Plan in Piped Water Supply , which identified uncovered villages that were technically suitable for piped-water

but were flagged as potentially creating licensing conflicts (3i, 2020). This research will compare the existing, license-constrained allocation with an optimal network benchmark to determine whether current territorial assignments increase pipe length, investment costs per household, or delays in last-mile coverage.

### c. Proposed plan and expected results

The analysis proceeds in two parts. The first part identifies whether licensing territories shaped by administrative boundaries create spatial misallocation by preventing villages from being served by their lowest cost feasible providers, leaving them uncovered even when a nearby network across the boundary could extend service at lower cost. The second part constructs an optimal benchmark, evaluates how far the observed allocation departs from this benchmark and quantifies the potential gains from alternative assignments. The expected data are listed in Table 1.

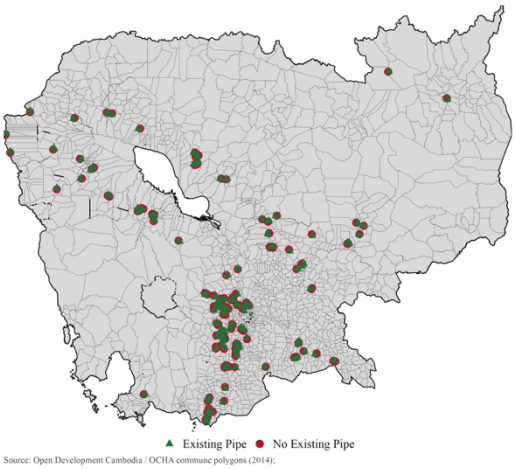


Figure 2 Piped Water Coverage Across Commune Boundaries.

| Data Type | Source |
|---|---|
| Water-supplier licenses | Water Supplier Monitoring System |
| Administrative boundaries | Open Development Cambodia |
| Piped-water maps | Provincial Investment Plan in Piped Water Supply Study |
| Household water use | Cambodia Socio-Economic Survey |

Table 1 Data Source.

**Identifying spatial misallocation.** The first part uses a spatial difference in discontinuities design to compare villages near the same predetermined administrative boundary before and after licensing begins to shape network expansion. As illustrated in Figure 3, the strategy exploits the fact that administrative boundaries determine operators' exclusive service territories while remaining unrelated to underlying engineering conditions. Before licensing shapes network decisions, villages near the boundary should have similar water access and comparable geographic determinants of connection costs. After licensing, however, if territorial assignments restrict network expansion across boundaries, water access may exhibit a discontinuity at the border. The analysis estimates whether such a discontinuity emerges after licensing relative to the pre-licensing period and examines its effects on water access and household welfare. A post-licensing jump in these outcomes would provide evidence that territorial allocation, rather than physical feasibility, contributes to spatial misallocation.

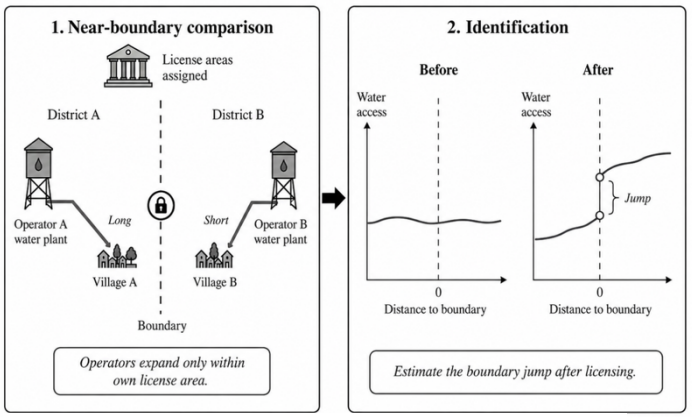


Figure 3 Illustration of the Identification Strategy.

**Evaluating alternative allocations**. The second part evaluates how much efficiency could be improved under alternative licensing arrangements. Using village level panel data on license assignments and subsequent network expansion, the study will characterize how operators expand their networks under the existing allocation system. Based on this observed expansion behavior, I will develop a model to predict how operators would respond under alternative territorial assignments. The model will then simulate counterfactual allocations, such as assigning service territories based on engineering costs rather than administrative boundaries, and quantify whether such alternatives could reduce expansion costs and improve overall social welfare.

### d. Expected impacts

This study will generate insights for Cambodian policymakers and JICA on territorial licensing systems that balance private investment incentives and efficient infrastructure allocation. By assessing social costs of misaligned territories and alternative allocation rules, it may inform discussions of Cambodian reforms and JICA-supported efforts to expand networks toward universal access at lower cost. Through policy dialogue, exchanges, and conferences, I will share these insights with Cambodian and Japanese policymakers. Academically, it will advance infrastructure-regulation research by showing how operating rights shape network-expansion efficiency.